

Abdullah Khalid Fadillah

AI/ML Engineer | Full-Stack Developer | IoT & Embedded Systems

Pontianak, West Kalimantan, Indonesia | abdullah20khalid20@gmail.com | Open to remote work

[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [Hugging Face](#)

SUMMARY

Computer Systems Engineer who builds AI systems end to end, from sensors to predictive models to full-stack dashboards. Hands-on with LLM fine-tuning (QLoRA), RAG, computer vision, and time-series forecasting, delivered as working web apps on FastAPI, React, and PostgreSQL. Published ML models on Hugging Face. Available for remote contract and startup roles.

TECHNICAL SKILLS

AI / ML: LLM Fine-tuning (QLoRA), RAG, Computer Vision (YOLOv8), NLP (IndoBERT), LSTM Time-Series, TensorFlow, Keras, scikit-learn

Full-Stack: Python, FastAPI, React, Laravel, PostgreSQL, Supabase, MySQL, REST API, JavaScript, TypeScript, Tailwind CSS

IoT / Embedded: ESP32, ESP8266, AVR/ATmega, MQTT, LoRa, RS485/Modbus, PCB Design (EasyEDA)

Tools / DevOps: Git, Docker, Linux/Ubuntu VPS, Nginx, Vercel, Hugging Face Spaces, Google Colab

EXPERIENCE

Web Developer, Internship

PT Perkebunan Nusantara IV (PTPN IV) Regional V, Pontianak · 2024

- Built a web-based procurement system (Laravel, MySQL, REST API) covering procurement requests, stock management, and reporting.
- Digitized a manual spreadsheet-based workflow, shortening processing time and improving stock-data accuracy.
- Coordinated with users across departments; this domain experience later informed the Deschain procurement platform.

SELECTED PROJECTS

Deschain — AI Collective Procurement Platform for MSMEs

Team Lead · 2026

- RAG consultation engine with a 4-layer fallback (Groq / Claude / Ollama / template) and a fine-tuned Mistral-7B (QLoRA), published as [joezy99/deschain-umkm-7b](#) on Hugging Face.
- React + FastAPI + Supabase; 15-page frontend and 27 API endpoints. Scored 100/100 on the compliance audit and reached Top 480 at PIDI-DIGDAYA X Hackathon 2026 (BI, OJK, AFTECH).

Real-Time Human Detection — YOLOv8 + ESP32-CAM

2026

- Custom-trained YOLOv8n model running real-time inference on an ESP32-CAM MJPEG stream, served through a Flask web app that streams annotated detections to the browser.
- Tech: YOLOv8, Python, Flask, OpenCV, ESP32-CAM, MQTT.

MATAHARI — IoT + LSTM Solar Radiation Forecasting

Undergraduate Thesis · 2026

- End-to-end pipeline: pyranometer (RS485/Modbus) to ESP32 to MQTT to an LSTM forecast, deployed on an Ubuntu VPS (Mosquitto, MySQL, Nginx, Flask API).
- LSTM achieved RMSE 31.68 and R2 0.9847 on a 15-minute horizon; MQTT Quality of Service evaluated with the TIPHON standard.

Indonesian Sentiment Analysis — NLP

2026

- Denoised noisy star-rating labels via star-rating x InSet-lexicon cross-validation, reaching 92%+ accuracy across SVM, Logistic Regression, and IndoBERT.

EDUCATION

B.Sc. in Computer Systems Engineering (S.Kom)

Universitas Tanjungpura, Pontianak · Expected 2026

Concentration: Network Intelligence Communication (NIC)

CERTIFICATIONS & ACHIEVEMENTS

- Dicoding Academy (2026): Building ML Systems, Fundamental Deep Learning, ML for Beginners, Data Science Fundamentals, AI Fundamentals, Prompt Engineering, Python Programming, Project Management.
- Top 480 & 100/100 audit — PIDI-DIGDAYA X Hackathon 2026 (Team Lead). Semifinalist, Student Category — BI-OJK Hackathon 2025.

LANGUAGES

Indonesian (Native) · English (B1-B2, TUTEF 508)